A CATEGORICAL GENERALIZATION FOR THOMPSON'S GROUPS

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Abstract. We propose a generalization of Thompson's groups obtained via the notion of a crossed $\mathcal{C}$-group for a small category $\mathcal{C}$. When $\mathcal{C}$ is fixed to be the so-called category of binary trees $\mathcal{T}$, our generalization subsumes Zaremsky's groups defined by cloning systems. In this case we also provide an algebraic classification theorem for crossed $\mathcal{T}$-groups (and, in particular, for generalized $\mathcal{T}$-Thompson's groups). We prove that the class of $\mathcal{C}$-generalized Thompson's groups with $\mathcal{C}$ not fixed is closed under taking subgroups.

Introduction

Thompson's groups possess remarkably rich algebraic properties, and numerous generalizations have appeared in the literature. In this note we explain how they are related to a different notion of a crossed group over a category. Intuitively, a crossed group over category $\mathcal{C}$ is a category $\mathcal{C}G$ with $\mathcal{C}$ as a wide subcategory such that any morphism in $\mathcal{C}G$ is uniquely factored as a morphism in $\mathcal{C}$ followed by an automorphism. For a fixed connected $\mathcal{C}$ we consider different crossed groups $\mathcal{C}G$. Its localization groupoid $\mathsf{Loc}\ \mathcal{C}G$ is equivalent to some group. In this note we explain how the groups that we get resemble the Thompson's groups.

One of the natural questions on crossed groups is the question of determining the homotopy type of their realizations $|\mathcal{C}G|$. If $\mathcal{C}$ is an arbitrary small connected category, the homotopy type can be anything. For example, if $\mathcal{C} = \Delta$ is the simplex category, then crossed Δ groups are the same as crossed simplicial groups that have been studied in [**?**, **?**, **?**]. In particular, a simplicial group is a crossed simplicial group. In [**?**] for a crossed simplicial group G the authors used Quillen's theorem B to provide the fibration sequence

$$|G_*| \to |\Delta| \to |\Delta G|$$

thus, $\pi_i(\Delta G) = \pi_{i-1}(|G_*|)$. Since simplicial group model loop spaces, ΔG can have any homotopy type.

We focus on a simpler case, where the category $\mathcal{C}G$ admits a calculus of fractions. In this case, by Theorem [**?**] the homotopy type of $\mathcal{C}G$ is that of the realization of its fundamental group $B\,\pi_1\,\mathsf{Loc}\ \mathcal{C}G$. Thus the problem reduces to the study of the groupoid localization of the category $\mathcal{C}G$.

Overview of results. First, we define crossed groups and study their basic properties. Then we prove the theorem on subgroups of them. Then we study thompson's groups. Then proofs.

Notable examples include those developed in [**?**, **?**, **?**, **?**]. Each of these constructions employs Ore categories and their localizations as the principal framework for constructing the corresponding groups.

In this note, we propose an additional generalization, which also makes essential use of Ore categories and their localizations. Although our approach is conceptually related to the perspectives of [**?**, **?**], we aim to present it from a distinct viewpoint that may clarify certain algebraic properties of these groups. In particular, our approach allows us to prove that the proposed class of generalized Thompson's groups is closed under taking subgroups and provide a structural theorem. For a standard treatment of original Thompson groups F, T, and V, we refer the reader to [**?**].

To preserve the continuity of exposition, detailed computational proofs are collected at the end of the paper.

Overview of results. For a small sufficiently good[1] small category $\mathcal{C}$ with a fixed object $c \in \mathcal{C}$ we consider its category of crossed $\mathcal{C}$-groups $\mathsf{Crossed}(\mathcal{C})$. Each crossed $\mathcal{C}$-group G comes equipped with a small category $\mathcal{C}G$ containing $\mathcal{C}$ as a wide subcategory. We introduce a full subcategory of $\mathsf{Crossed}(\mathcal{C})$ called $\mathsf{GOCrossed}(\mathcal{C})$ consisting of so-called injective crossed $\mathcal{C}$-groups. We prove that G is an injective crossed $\mathcal{C}$-group iff its category $\mathcal{C}G$ is an Ore category. Thus, for each $G \in \mathsf{GOCrossed}(\mathcal{C})$ there is a a well-behaved localization groupoid $\mathsf{Loc}\ (\mathcal{C}G)$.

Definition. For an injective crossed $\mathcal{C}$-group G its generalized $\mathcal{C}$-Thompson's is the group

$$F_{\mathcal{C}}(G) := \mathrm{Aut}_{\mathsf{Loc}\ (\mathcal{C}G)}(c) = \pi_1(\mathsf{Loc}\ (\mathcal{C}G), c).$$

Next, we introduce the so-called category of binary trees $\mathcal{T}$. We utilize this category by showing that all classical Thompson's groups are generalized $\mathcal{T}$-Thompson's groups. In particular, F, T, V and the braided Thompson's-Dehornoy group bV are examples of generalized $\mathcal{T}$-Thompson's groups. We formalize this in Lemma 4.7 proving that each cloning system in the sense of M.Zaremsky gives rise to a crossed $\mathcal{T}$-group in a canonical way. For generalized $\mathcal{T}$-Thomspon's groups we prove an algebraic structural theorem.

Theorem (Theorem 5.1)**.** *There exists a $\mathcal{T}$-crossed group H such that for any crossed $\mathcal{T}$-group G there is a canonical exact sequence of crossed $\mathcal{T}$-groups of the form*

$$1 \to \mathcal{T}P \to \mathcal{T}G \to \mathcal{T}H,$$

where $P \subseteq G$ is a genuine $\mathcal{T}$-group.

This result can be translated to give an algebraic classification of generalized $\mathcal{T}$-groups.

Theorem (Theorem 5.2)**.** *Let $\mathcal{F}_{\mathcal{T}}(G)$ be a generalized $\mathcal{T}$-Thompson's group, then there exists a natural short exact sequence of generalized $\mathcal{T}$-Thompson's groups:*

$$1 \to \mathrm{colim}_{t \in \mathcal{T}_0} N_t \to \mathcal{F}_{\mathcal{T}}(G) \to \mathcal{F}_{\mathcal{T}}(H),$$

where N is a genuine $\mathcal{T}$-group.

Next, we prove that a subgroup of a generalized Thompson's group is also a generalized Thompson's group. We deliberately omit mentioning the base category above, since the correct formulation is the following.

Theorem (Theorem 3.1)**.** *Let $\mathcal{F}_{\mathcal{C}}(G)$ be a generalized $\mathcal{C}$-Thomspon's group and let $H \leq \mathcal{F}_{\mathcal{C}}(G)$ be its subgroup. Then there exists a sufficiently good category $\mathcal{Q}$ and an injective $\mathcal{Q}$-group N so that*

$$H \cong \mathcal{F}_{\mathcal{Q}}(N).$$

The proof of this theorem relies on the theory of groupoid coverings.

Ackowledgments. The author would like to express sincere gratitude to Andrey Ryabichev for drawing the beautiful pictures for this paper.

(1) Overview, intro
(2) Detailed overview of the case of Thompson's groups.
(3) Rewrite from gaunt to thin.
(4) crossed groups have coproducts (add a proof).
(5) rewrite the proof using Vasya's ideas.
(6) write about framed confs of manifolds, unframed confs of torus
(7) strange idea: take G to be the generalized Thompson's group corresponding to homotopy braids. what is this group? is Malyutin's quasimorphism non-trivial on it? How is it related to the question of amenability of F?
(8) add vasya ionin to Ackowledgments.

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[1]The conditions imposed on $\mathcal{C}$ to be sufficiently good are: $\mathcal{C}$ needs to be connected, Ore and gaunt (without non-trivial automorphisms). We discuss these conditions in detail below.

(9) reference for Calculus of fractions $\Rightarrow$ 1-truncated.
(10) fractals (what Davide told me)?????

1. Crossed groups

1.1. Definitions. We adopt the convention of right-to-left composition throughout this article; that is, for morphisms $\bullet \xrightarrow{f} \bullet \xrightarrow{g} \bullet$, we denote the composition by fg.

For a category $\mathcal{C}$, we denote the hom-sets $\hom_{\mathcal{C}}(a,b)$ by $\mathcal{C}(a,b)$. When $\mathcal{C}$ is a small category and $\mathcal{D}$ is any category, the functors $\mathcal{C} \to \mathcal{D}$ form a category whose objects we call $\mathcal{C}$-objects in $\mathcal{D}$. This allows us to speak of $\mathcal{C}$-groups, $\mathcal{C}$-spaces, and so forth. Crossed simplicial groups were introduced by Fiedorowicz and Loday [**?**] and, under the name skew-simplicial groups, by Krasauskas [**?**]. Berger and Moerdijk [**?**] generalized this notion to that of a crossed $\mathcal{C}$-group, where $\mathcal{C}$ is a small category. We present two equivalent definitions of crossed $\mathcal{C}$-groups below. Since crossed $\mathcal{C}$-groups generalize $\mathcal{C}$-groups, we sometimes refer to the latter as *genuine* $\mathcal{C}$-groups for emphasis.

A category $\mathcal{C}$ is called *gaunt* if it has no nontrivial isomorphisms. For the remainder of this section, we fix a small, gaunt, connected category $\mathcal{C}$.

Definition 1.1. A *crossed $\mathcal{C}$-group* consists of a collection of groups $\{G_c\}_{c\in\mathcal{C}}$ together with a small category $\mathcal{C}G$ satisfying:

(1) $\mathrm{Ob}(\mathcal{C}G) = \mathrm{Ob}(\mathcal{C})$,
(2) $\mathcal{C}$ is a subcategory of $\mathcal{C}G$,
(3) $\mathrm{Aut}_{\mathcal{C}G}(c) = G_c^{\mathrm{op}}$ for all $c \in \mathcal{C}$,
(4) Every morphism in $\mathcal{C}G$ factors uniquely as $\varphi \cdot g$, where $\varphi \in \mathcal{C}(c_1, c_2)$ and $g \in G_{c_2}$.

The category $\mathcal{C}G$ is called the *category of the crossed $\mathcal{C}$-group G*.

Let G be a crossed $\mathcal{C}$-group, then for any objects $a, b \in \mathcal{C}$ and any morphisms $f \in \mathcal{C}(a,b)$, $g \in G_a$, there exist unique morphisms $g^*(f) \in \mathcal{C}(a,b)$ and $f^*(g) \in G_b$ such that

$$g \cdot f = g^*(f) \cdot f^*(g).$$

Lemma 1.2. *The assignment $f^* : G_{\mathrm{dom}\, f} \to G_{\mathrm{cod}\, f}$ defines a $\mathcal{C}$-set structure. Furthermore, if all the maps g^* are trivial (that is, $g^* = \mathsf{id}$), then this structure is an genuine $\mathcal{C}$-group.* □

The lemma above motivates the following equivalent definition, which more closely resembles that of [**?**].

Definition 1.3. A *crossed $\mathcal{C}$-group* is a functor $G : \mathcal{C}^{\mathrm{op}} \to \mathsf{Set}$ equipped with:

(1) A group structure on G_c for each $c \in \mathcal{C}$,
(2) A left G_c-action on $\mathcal{C}(c, s)$ for every $s \in \mathcal{C}$,

satisfying the following axioms for all composable morphisms and group elements:

(1) $g^*(fh) = g^*(f) \cdot (f^*(g))^*(h)$,
(2) $g^*(\mathrm{id}_c) = \mathrm{id}_c$,
(3) $f^*(hg) = (g^*(f))^*(h) \cdot f^*(g)$,
(4) $f^*(1_{\mathrm{dom}\, f}) = 1_{\mathrm{cod}\, f}$.

A morphism between crossed $\mathcal{C}$-groups G_* and H_* is a functor $\mathcal{C}G \to \mathcal{C}H$ that is the identity on the subcategory $\mathcal{C}$. This defines the category $\mathsf{Crossed}(\mathcal{C})$ of crossed $\mathcal{C}$-groups. The following lemma is useful.

Lemma 1.4. *The category $\mathsf{Crossed}(\mathcal{C})$ of crossed $\mathcal{C}$-groups*

(1) *is closed under pullbacks of monomorphisms;*
(2) *has a zero object (i.e. an object which is both initial and terminal).*
(3) *has all coproducts.*

□

2. General knowledge on groupoids and localizations

The full subcategory of groupoids Gpd is reflective in Cat; that is, the inclusion $\mathsf{Gpd} \to \mathsf{Cat}$ has a left adjoint $\mathsf{Loc}\colon \mathsf{Cat} \to \mathsf{Gpd}$. Denote as $\mathsf{loc}\ : \mathsf{Id}_{\mathsf{Cat}} \Rightarrow \mathsf{Loc}$ the corresponding coaugmentation natural transformation.

Definition 2.1. a map is isofibration if

For example, epimorphisms of crossed groups are isofibrations.

Definition 2.2. a map is cover if

Definition 2.3. admits a calculus of fractions

For certain categories $\mathcal{C}$, the localization admits a simple description via calculus of fractions. These are known as *Ore categories*.

Definition 2.4. We say that a small category $\mathcal{C}$ is an *Ore category* if following properties hold:

(1) $\mathcal{C}$ is cancellative, meaning that for all morphisms p, q, r for which the following expressions are defined, we have:

$$pq = rq \Rightarrow p = r, \tag{O1}$$

$$qp = qr \Rightarrow p = r; \tag{O1}$$

(2) $\mathcal{C}$ has common right multiples. That is, given two morphisms p, q with the same domain, there exist morphisms r, s such that $pr = qs$:

$$\begin{tikzcd} & \bullet \arrow[dr, dashed, "\exists\, r"] & \\ \bullet \arrow[ur, "p"] \arrow[dr, "q"'] & & \bullet\ . \\ & \bullet \arrow[ur, dashed, "\exists\, s"'] & \end{tikzcd} \tag{O2}$$

The following theorem is a classical result in this area (see, e.g., [**?**]).

Theorem 2.5. *The following two conditions are equivalent:*

(1) $\mathcal{C}$ *is an Ore category.*
(2) *There is the following description of the objects and morphisms in* $\mathsf{Loc}\ \mathcal{C}$*:*
   (a) *The objects of* $\mathsf{Loc}\ \mathcal{C}$ *are the same as the objects of* $\mathcal{C}$.
   (b) *Every morphism in* $\mathsf{Loc}\ \mathcal{C}$ *can be represented as a composition* pq^{-1} *(where* $p, q \in \mathsf{Arr}(\mathcal{C})$ *with the same codomain).*

Morphisms of the form pq^{-1} in the category $\mathsf{Loc}\ \mathcal{C}$ will be referred to as *formal fractions*.

2.1. Injectivity condition on crossed groups. It is obvious that for for the crossed $\mathcal{C}$ group G to have the category $\mathcal{C}G$ satisfy Ore conditions it is necessary for $\mathcal{C}$ to satisfy them also. To formulate the sufficient and necessary conditions for the Ore properties to be satisfied by $\mathcal{C}G$ we introduce a new notion.

Definition 2.6. We say that a crossed $\mathcal{C}$-group G is *injective* if for every morphism $f \in \mathsf{Arr}(\mathcal{C})$, its action $f^*\colon G_{\mathrm{dom} f} \to G_{\mathrm{cod} f}$, defined by the presheaf structure, is an injective function.

In the appendix we prove tha the injectivity condition fully characterizes those G with Ore $\mathcal{C}G$. That is, the following lemma holds.

Lemma 2.7. *Let* $\mathcal{C}$ *be a small category and* G *a crossed group over it. The following conditions are equivalent:*

(1) *The category* $\mathcal{C}G$ *is an Ore category;*
(2) $\mathcal{C}$ *is an Ore category, and* G *is injective.*

We organize the introduced classes of categories and their crossed groups into full subcategories of corresponding categories.

Definition 2.8. (1) Define GOCat to be the full subcategory of Cat whose objects are gaunt, connected Ore categories.
(2) For a fixed $\mathcal{C} \in \mathsf{GOCat}$, define $\mathsf{GOCrossed}(\mathcal{C})$ to be the full subcategory of $\mathsf{Crossed}(\mathcal{C})$ of all injective crossed groups.
(3) Define $\mathsf{GOCrossed}$ as the full subcategory of $\mathsf{Crossed}$ consisting of pairs $(\mathcal{C}, G)$ such that $\mathcal{C} \in \mathsf{GOCat}$ and $G \in \mathsf{GOCrossed}(\mathcal{C})$.

2.2. Generalized Thompson's groups. In this section we assume that all categories and groupoids considered are given a fixed object, i.e. they are punctured categories and groupoids. If not stated otherwise, we denote the fixed object simply as $* \in \mathsf{Obj}(\mathcal{C})$. If $\mathcal{E}$ is some category of categories, e.g. $\mathcal{E} = \mathsf{Gpd}$ the category of groupoids or $\mathcal{E} = \mathsf{GOCat}$, then put $\mathcal{E}_*$ to be the category of punctured $\mathcal{E}$-categories with functors preserving the fixed object. Note that we consider the category $\mathcal{T}$ to have $1 \in \mathcal{T}$ as the fixed object. If $c \in \mathsf{Obj}(\mathcal{C})$, then denote as $\pi_1(\mathcal{C}, c)$ the group of automorphisms $\mathrm{Aut}_{\mathcal{C}}(c)$. If we consider punctured categories, then it turns π_1 into a functor $\pi_1 : \mathsf{Cat}_* \to \mathsf{Grp}$.

Definition 2.9 (A generalized $\mathcal{C}$-Thompson's group)**.** Let $\mathcal{C}$ be a category in GOCat_* and let $G \in \mathsf{GOCrossed}(\mathcal{C})$, define its generalized $\mathcal{C}$-Thompson's group as

$$\mathcal{F}_{\mathcal{C}}(G) = \pi_1(\mathsf{Loc}\ \mathcal{C}G).$$

We state here a number of examples for $\mathcal{C} = \mathcal{T}$.

(1) The Thompson's group F is isomorphic to $\mathcal{F}_{\mathcal{T}}(\{1\}_{n\geq 1})$;
(2) The Thompson's group T is isomorphic to $\mathcal{F}_{\mathcal{T}}(\{\mathbb{Z}/n\}_{n\geq 1})$;
(3) The Thompson's group V is isomorphic to $\mathcal{F}_{\mathcal{T}}(\{\Sigma_n\}_{n\geq 1})$;
(4) The Dehornoy's braided Thompson's group bV is isomorphic to $\mathcal{F}_{\mathcal{T}}(\{B_n\}_{n\geq 1})$.

We start the investigation of the properties of generalized Thompson's groups with simple observations. First, we explain how one can think about elements of $\mathcal{F}_{\mathcal{C}}(G)$. Recall, that in the "classical" Thompson's groups (e.g. given by cloning systems) each element can be presented as a pair of trees and some group element "between" those trees. In our generalized setting, something similar holds.

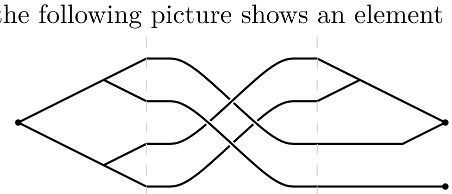
Lemma 2.10. *An element of $(\mathsf{Loc}\ \mathcal{C}G)(c_1, c_2)$ can be encoded as a triple (t_1, g, t_2) where $t_1 \colon c_1 \to c, t_2 \colon c \to c_2$, $g \in G_c$ and c is some object of $\mathcal{C}$.*

Proof. Each morphism in $(\mathsf{Loc}\ \mathcal{C}G)$ from c_1 to c_2 is a formal fraction of some $f \in \mathcal{C}G(c_1, c)$ and $g \in \mathcal{C}G(c_2, c)$. Then, by definition $f = f_{\mathcal{C}} \cdot f_G$ and $g = g_{\mathcal{C}} \cdot g_G$, where $f_{\mathcal{C}}, g_{\mathcal{C}}$ are morphisms in $\mathcal{C}$ and f_G, g_G are elements of G_c. The following simple computation provides the desired encoding as a triple.

$$fg^{-1} = f_{\mathcal{C}}(f_G g_G^{-1})g_{\mathcal{C}}^{-1}.$$

□

Remark 2.11. For example, the following picture shows an element of $(\mathsf{Loc}\ \mathcal{T}B)(1, 2)$:



Note that by remark above the elements of $\mathcal{F}_{\mathcal{C}}(G)$ are triples (t_1, g, t_2), where $t_1, t_2 \in */\mathcal{C}$ with the same codomain and $g \in G_{\mathrm{cod}\ t_1}$. In the case $\mathcal{C} = \mathcal{T}$ the elements of $1/\mathcal{T}$ are encoded via binary trees. As in the classical case, we provide a necessary and sufficient conditions on a triple (t_1, g, t_2) to present the trivial element of $\mathcal{F}_{\mathcal{C}}(G)$.

Lemma 2.12 (Word problem for the normal form)**.** *Let $\mathcal{F}_{\mathcal{C}}(G)$ be a generalized $\mathcal{C}$-Thompson's group. Suppose that $(t_1, g, t_2) \in \mathcal{F}_{\mathcal{C}}(G)$ is equal to 1 in the group, then $t_1 = t_2$ and $g = 1$.*

The following two lemmas follow easily from 2.12.

Lemma 2.13. *The map induced from the initial crossed group $\mathcal{F}_{\mathcal{C}}(1) \hookrightarrow \mathcal{F}_{\mathcal{C}}(G)$ is a monomorphism of groups.* □

Lemma 2.14. *For each $t \in */\mathcal{C}$ the map $G_{\mathrm{cod}\, t} \to \mathcal{F}_{\mathcal{C}}(G)$ given by $g \mapsto (t, g, t)$ is a monomorphism of groups.* □

3. Closed under subgroups

In this section we prove the following result: the class of all generalized Thompson's groups (without a fixed base category) is closed under subgroups. That is, we prove the following theorem.

Theorem 3.1. *Let $H \leq \mathcal{F}_{\mathcal{C}}(G)$ be a subgroup of a generalized Thompson's group. Then there exist $\mathcal{D} \in \mathsf{GOCat}_*, N \in \mathsf{GOCrossed}(\mathcal{D})$ such that $H = \mathcal{F}_{\mathcal{D}}(N)$. Also, the construction of $\mathcal{D}$ and N is functorial.*

For the proof we use a reformulation of the Galois correspondence principle in the groupoid language (see [**?**]). For the exposition to be self-contained, we remind here the basics of this theory.

Definition 3.2. Let $\mathcal{E}, \mathcal{B}$ be two small groupoids. A cover $p\colon \mathcal{E} \to \mathcal{B}$ is a functor such that

(1) It is surjective on objects;
(2) It is bijective on the slice-categories $x/\mathcal{E}$ for all $x \in \mathsf{Obj}(\mathcal{E})$.

All covers of a fixed groupoid $\mathcal{B}$ form a category, which we denote $\mathsf{Cov}(\mathcal{B})$. In order to formulate the Galois correspondence, we need to introduce the category $\mathsf{Subgroup}\, G$ for a group G. Put

(1) Objects of $\mathsf{Subgroup}\, G$ to be the subgroups of G;
(2) Morphisms from $H \leq G$ to $N \leq G$ to be the set $\{g \in G \;:\; g^{-1}Hg \subset N\}$ with the obvious composition law.

Theorem 3.3 (Galois correspondence theorem, ch. 3 in [**?**])**.** *Let Γ be a small connected groupoid with a fixed object. There is a natural in Γ equivalence of categories:*

$$\tilde{\mathcal{E}}\colon \mathsf{Subgroup}\, G \longleftrightarrow \mathsf{Cov}(\Gamma) : \pi_1$$

Proof of Theorem 3.1. Let $H \leq \mathcal{F}_{\mathcal{C}}(G)$ be a subgroup. By Galois correspondence theorem 3.3 we obtain a group covering $\mathcal{E} \to \mathsf{Loc}\; \mathcal{C}G$. Consider the following pullback diagram.

$$\begin{tikzcd}
\mathcal{Q} \arrow[r] \arrow[d] \arrow[dr, phantom, very near start, "\lrcorner"] & X \arrow[r] \arrow[d] \arrow[dr, phantom, very near start, "\lrcorner"] & \mathcal{E} \arrow[d, "\text{cover}"] \\
\mathcal{C} \arrow[r] & \mathcal{C}G \arrow[r] & \mathrm{Loc}\, \mathcal{C}G
\end{tikzcd}$$

We utilize the following technical statements that we prove in the appendix.

Lemma 3.4. *Let $\mathcal{C} \to \mathcal{B}$ be an inclusion of an Ore category to a groupoid and let $p\colon \mathcal{E} \to \mathcal{B}$ be a groupoid covering. Then the pullback $\mathcal{E} \times_{\mathcal{B}} \mathcal{C}$ is an Ore category.*

Lemma 3.5. *Let $\mathcal{C}$ be a connected Ore category, let $p\colon \mathcal{E} \to \mathsf{Loc}\; \mathcal{C}$ be a groupoid covering. Then the structural map $\pi_{\mathcal{E}}\colon \mathcal{C} \times_{\mathsf{Loc}\, \mathcal{C}} \mathcal{E} \to \mathcal{E}$ induces an equivalence of categories*

$$\mathsf{Loc}\, (\mathcal{C} \times_{\mathsf{Loc}\, \mathcal{C}} \mathcal{E}) \to \mathcal{E}.$$

Lemma 3.6. *In the diagram above*

(1) *The category $\mathcal{Q}$ is a connected gaunt Ore category,*
(2) *The functor $\mathcal{Q} \to X$ is an inclusion of a GOCat category into its crossed group category.*

Using Lemma 3.6 we conclude that $X \cong \mathcal{Q}N_*$ for some crossed $\mathcal{Q}$-group N_*. Using Lemma 3.5 we see that $E \cong \mathsf{Loc}\; \mathcal{Q}N_*$. Hence, $H = \pi_1(\mathsf{Loc}\; \mathcal{Q}N_*) =: \mathcal{F}_{\mathcal{Q}}(N)$. By the fact that the construction is a pullback, $\mathcal{Q}$ and N are functorial.

4. Generalizing Thompson's groups

This section presents the main application of the theory outlined above.

We now introduce our main example of a small gaunt category $\mathcal{T}$, which will be used to construct the standard Thompson's groups F, T, and V. We call it the *category of binary trees*.

Definition 4.1. The category $\mathcal{T}$ is defined by the following presentation:

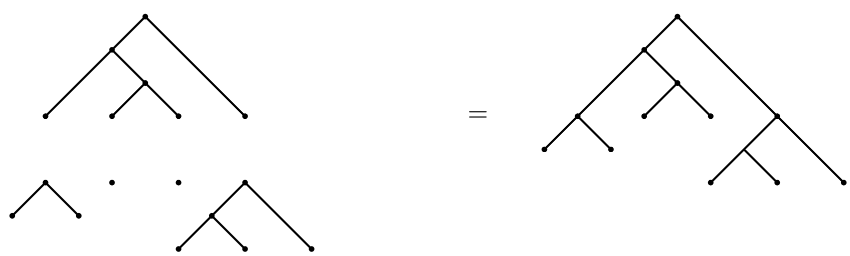
(1) the objects are natural numbers $1, 2, \dots$;
(2) for each n, there are n morphisms $\mathcal{T}(n, n+1)$, denoted $e_1^{(n)}, \dots, e_n^{(n)}$ (superscripts will often be omitted);
(3) these morphisms satisfy the relations:

$$e_j e_i = e_i e_{j+1}, \quad \text{for } i < j.$$

Remark 4.2. The category $\mathcal{T}$ admits the following combinatorial description (see [?]): morphisms $\mathcal{T}(n, m)$ correspond bijectively to ordered n-tuples of rooted binary trees with a total of m leaves. Composition $\mathcal{T}(n, m) \times \mathcal{T}(m, k) \to \mathcal{T}(n, k)$ corresponds to concatenation of trees.

Remark 4.3. The given relations resemble the standard simplicial identities. Indeed, if we add the relations $e_i e_i = e_i e_{i+1}$ and delete the object 1, the resulting category would be equivalent to Δ_{inj}.

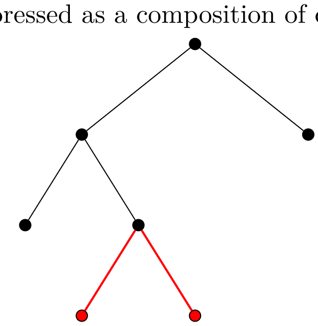
The figure below depicts morphisms $1 \to 4$ (top left) and $4 \to 7$ (bottom left). Their composition, a morphism $1 \to 7$ is shown on the right.



Sometimes we work with the slice category $\mathcal{T}_0 := 1/\mathcal{T}$. From the tree description, the objects of $\mathcal{T}_0$ are all rooted binary trees, and $\mathcal{T}_0$ is thin (i.e., a preorder). Alternatively, the categorical structure of $\mathcal{T}_0$ is determined by the inclusion of trees as rooted subtrees.

Let $f\colon 1 \to n$ be an object of $\mathcal{T}_0$. Multiplication of f by $e_i\colon n \to n+1$ is called *caret addition*. The subtree added to f in this process is called a *caret*. The figure below shows a red caret added to the tree $1 \to 3$ (black), resulting in the tree $1 \to 4$.

Every morphism in $\mathcal{T}_0$ can be expressed as a composition of caret additions.



Crossed $\mathcal{T}$-groups. Specializing to the case $\mathcal{C} = \mathcal{T}$, we find that a crossed $\mathcal{T}$-group is specified by a sequence of groups $\{G_n\}_{n \geq 1}$ together with maps

$$e_i : G_n \longrightarrow G_{n+1}, \quad \text{for } i = 1, \dots, n,$$

satisfying certain relations.

Since every morphism in $\mathcal{T}G$ can be uniquely written as $t \cdot g$ with $t \in \mathcal{T}$ and $g \in G_*$, one may visualize such a morphism as the tree diagram representing t decorated by the group element g. This provides a useful intuition, though it is not a formal definition.

We now present several examples of crossed $\mathcal{T}$-groups. It is explained in Lemma 4.5 that these are indeed crossed $\mathcal{T}$ groups.

4.0.1. *The braided crossed $\mathcal{T}$-group.* This example, while not the simplest, is particularly instructive for our purposes.

Let B_n denote the Artin braid group on n strands. For each n, there are n *cabling maps*

$$e_i : B_n \to B_{n+1}, \quad i = 1, \dots, n.$$

The map e_i replaces the i-th strand with two parallel strands. These maps define the crossed $\mathcal{T}$-group B_* and its associated category $\mathcal{T}B_*$.

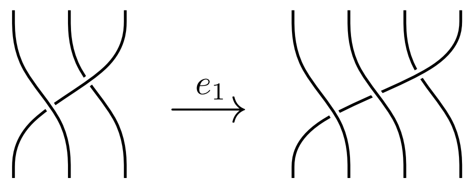


FIGURE 1. Example of the cabling operation

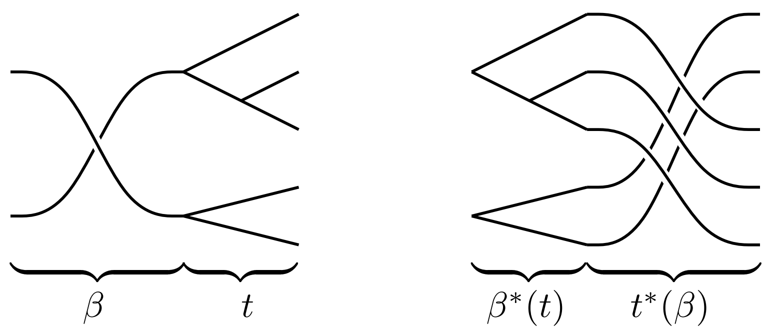


FIGURE 2. Example of a morphism $2 \to 5$ structure maps in $\mathcal{T}B$

4.0.2. *The hyperoctahedral crossed $\mathcal{T}$-group.* We now describe a specific crossed $\mathcal{T}$-group H_*. For each $n \geq 1$, define

$$H_n = (\mathbb{Z}/2)^n \rtimes \Sigma_n,$$

where the symmetric group Σ_n acts on $(\mathbb{Z}/2)^n$ by permuting coordinates. This group is commonly called the *hyperoctahedral group* or, in combinatorial contexts, the *group of signed permutations*.

We define maps $e_i \colon H_n \longrightarrow H_{n+1}$ for $1 \leq i \leq n$ by:

$$e_i((\varepsilon_1, \dots, \varepsilon_n), \sigma) = ((\varepsilon_1, \dots, \varepsilon_i, \varepsilon_i, \dots, \varepsilon_n), e_i^{\varepsilon_i}(\sigma)),$$

where:

- $e_i^0(\sigma)$ denotes the standard cabling operation for permutations, analogous to the braid group cabling,
- $e_i^1(\sigma)$ denotes *twisted cabling*, where the i-th strand is replaced by two intersecting strands, as illustrated in Figure 3.

Remark 4.4. Elements of H_n can be visualized as n strands with mod 2 twists, permuted according to σ. The maps e_i then correspond to inserting either parallel or intersecting strands at position i.

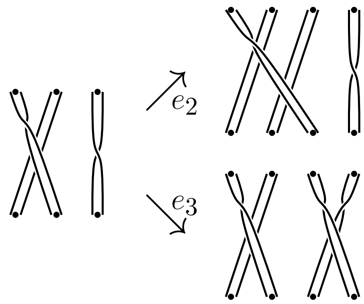


FIGURE 3. Multiplication by vertical concatenation and the action on $[n]$.

4.1. Connections with crossed simplicial groups. Now that we have defined injective $\mathcal{C}$-groups we can elaborate on the connections between crossed simplicial groups and crossed $\mathcal{T}$-groups, announced above.

By definition, a crossed Δ-group is a sequence of groups $G_0, G_1, G_2, \ldots$ together with morphisms $d_i : G_n \to G_{n-1}$ and $s_i : G_{n-1} \to G_n$ satisfying simplicial identities and crossed group properties. Note that identities in $\mathcal{T}$ between e_i's are all present in Δ as simplicial identities ivolving s_i's. Thus $\{G_i\}$'s also form a crossed $\mathcal{T}$-group. Also, there is the simplicial identity $s_j d_j = \mathsf{id}$ which implies that s_i are all injective functions. Thus we have proved the following lemma.

Lemma 4.5. *Any crossed simplicial group tautologically defines an injective crossed $\mathcal{T}$ group.*

4.2. Connections with cloning systems. Recall the definition of a cloning system due to M. Zaremsky and S. Witzel [**?**].

Definition 4.6. A cloning system consists of a directed system of groups (a functor $i \colon \mathbb{Z}_{\geq 1} \to \mathsf{Grp}$, where $i([n]) =: G_n$) together with:

(1) group homomorphisms $\rho_n \colon G_n \to \Sigma_n$ for each n, and
(2) set maps $\kappa_k^n \colon G_n \to G_{n+1}$ for $1 \leq k \leq n$, called *cloning functions*.

These data are required to satisfy a list of axioms (see [**?**]).

Given a cloning system $(i, \{\rho_n\}_{n\geq 1}, \{\kappa_k^n\}_{n\geq 1, 1\leq k\leq n})$, one associates a Thompson-like group whose elements are triples $[T_+, g, T_-]$, where $g \in G_n$ and T_+, T_- are rooted binary trees with n leaves (see [**?**] for the precise construction). For each cloning system there is a corresponding $\mathcal{T}$-crossed group. The following observation is an immediate unpacking of the definitions.

Lemma 4.7. *Let $(i, \{\rho_n\}_{n\geq 1}, \{\kappa_k^n\}_{n\geq 1, 1\leq k\leq n})$ be a cloning system. Then the assignment $n \mapsto G_n$ defines a crossed $\mathcal{T}$-group with e_i given by κ_i.*

It is worth noting that a system of structure maps $G_n \xrightarrow{i_{n,m}} G_m$ is not essential for defining a Thompson-like group. This was already observed by S. Witzel in [**?**], where an alternative formulation based on the *internal indirect product* of categories was introduced. The definition arising from crossed $\mathcal{T}$-groups is essentially equivalent to Witzel's formulation. However, our approach makes the connection with algebraic topology more transparent and facilitates results such as Theorem 3.1.

5. Structure theorem and its implications

Theorem 5.1 (Structure theorem, c.f. [**?**])**.** *Let G be a $\mathcal{T}$-crossed group, then there exists a natural exact sequence of $\mathcal{T}$-crossed groups.*

$$1 \to \mathcal{T}P \to \mathcal{T}G \to \mathcal{T}H,$$

where H is the hyperoctahedral crossed $\mathcal{T}$-group and P is an genuine $\mathcal{T}$-group.

Proof. This proof follows the proof of Theorem 3.6 of [?].

Consider the left action of G_n on $\mathcal{T}(n, n+1)$. Since $\mathcal{T}(n, n+1)$ is an n-element set, this defines a group homomorphism $\pi_S\colon G_n \to \Sigma_n$.

Note that by formulas from Definition 1.3 in a crossed $\mathcal{T}$-group one has

$$e_i^*(g_1 g_2) = (g_2^*(e_i))^*(g_1) \cdot e_i^*(g_2) = e_{g_2^{-1}(i)}(g_1) e_i(g_2).$$

It follows that e_i acts homomorphically on the kernel of π_S. However, this kernel can be not closed under the e_i-th still.

We will restrict the subgroup $\ker \pi_S$ to a smaller one so that it becomes closed under e_i's.

Define $\pi\colon G_n \to (\mathbb{Z}/2)^{\oplus n} \rtimes \Sigma_n$ as

$$g \mapsto \pi((\varepsilon_1, \ldots, \varepsilon_n), \pi_S(g)),$$

where $\varepsilon_i = 1$ if $e_i(g)$ acts on $\{i, i+1\}$ as a transposition and 0 otherwise.

Put $P_n := \ker \pi \trianglelefteq G_n$. We need to check that if $g \in P_n$, then $e_i(g) \in P_{n+1}$. It is straightforward that $e_i e_j(g)$ acts trivially on $\{1, \ldots, n+2\}$. □

Using the theorem 5.1 one can derive an analogous classification result for generalized $\mathcal{T}$-Thompson's groups.

Theorem 5.2. *Let $\mathcal{F}_{\mathcal{T}}(G)$ be a generalized $\mathcal{T}$-Thompson's group, then there exists a natural short exact sequence of generalized $\mathcal{T}$-Thompson's groups:*

$$1 \to \operatorname{colim}_{t \in \mathcal{T}_0} N_t \to \mathcal{F}_{\mathcal{T}}(G) \to \mathcal{F}_{\mathcal{T}}(H),$$

where N is a genuine $\mathcal{T}$-group.

Before we proceed with the proof we want to mention that for $G = \{B_n\}$ the theorem above specializes to a well known short exact sequence. Mainly, in this case the middle group is the braided Thompson's group $\mathcal{F}_{\mathcal{T}}(G) = bV$. Its projection to $\mathcal{F}_{\mathcal{T}}(H)$ has the Thompson's group V as its image. Its kernel has been studied from different perspectives. For example, in [?] it is called P_{br}.

Proof of Theorem 5.2. By writing $\mathcal{F}_{\mathcal{T}}(G)$ we implicitly assume that G is an injective crossed $\mathcal{T}$-group. Consider the exact sequence of crossed $\mathcal{T}$-groups given by 5.1:

$$1 \to \mathcal{T}N \to \mathcal{T}G \xrightarrow{\alpha} \mathcal{T}H.$$

The kernel $K = \ker\{\mathcal{F}_{\mathcal{T}}(G) \to \mathcal{F}_{\mathcal{T}}(N)\}$ consists of triples (t_1, g, t_2) such that $(t_1, \alpha(g), t_2) = 1$ in $\mathcal{F}_{\mathcal{T}}(N)$. Using Lemma 2.12 we obtain the following description:

$$K = \ker\{\mathcal{F}_{\mathcal{T}}(G) \to \mathcal{F}_{\mathcal{T}}(N)\} = \{(t, g, t) \mid t \in \mathcal{T}_0, g \in G_{|t|}, \alpha(g) = 1\}. \tag{5.1}$$

The map $\operatorname{colim}_{t \in \mathcal{T}_0} H_t \to K$ is given by $H_t \ni g \mapsto (t, g, t)$. Using the description 5.1 it is easy to see that it is an isomorphism. □

Appendix A. Routine verifications

To work with localizations, we will need the following useful lemma, which is a classical result in the theory of Ore categories (see [?]).

Lemma A.1. *Let $\mathcal{C}$ be an Ore category. Consider two formal fractions $p_1 q_1^{-1}$ and $p_2 q_2^{-1}$ in the localization. They are equal, i.e., $p_1 q_1^{-1} = p_2 q_2^{-1}$, if and only if there exist $r_1, r_2 \in \mathsf{Arr}(\mathcal{C})$ such that the* *following diagram commutes:*

$$\begin{tikzcd}
 & \bullet \arrow[dl, "p_1"'] \arrow[dr, "p_2"] & \\
\bullet \arrow[r, "r_1"] & \bullet & \bullet \arrow[l, "r_2"'] \\
 & \bullet \arrow[ul, "q_1"] \arrow[ur, "q_2"'] &
\end{tikzcd} .$$

Proof of Lemma 2.7.

Right Cancellability. For $a, b, c \in \mathcal{C}$ let $(\alpha, g), (\gamma, r) : b \to c$ and $(\beta, h) : \alpha \to \beta$ be morphisms such that $(\alpha, g) \cdot (\beta, h) = (\gamma, r) \cdot (\beta, h)$ or, unfolding the definition,

$$(\alpha \circ g_*(\beta),\ \beta^*(g) \cdot h) = (\gamma \circ r_*(\beta),\ \beta^*(r) \cdot h).$$

In the second coordinate by the injectiveness of β^* we get $g = r$ and then by the cancellability in $\mathcal{C}$ we get $\alpha = \gamma$.

Left Cancellability. Same argument, but one uses the fact that g_* is a bijection.

Common right multiples. For two arrow $(\alpha, g), (\gamma, r) \in \mathsf{Arr}(\mathcal{C}G_*)$, using existance of common right multiples property for $\mathcal{C}$ we get two arrows $A, B \in \mathsf{Arr}(\mathcal{C})$ such that $\alpha A = \gamma B$. Put $\beta := (g^{-1})_*(A), \delta := (r^{-1})_*(B)$ and $h := (\beta^*(g))^{-1}, p := (\delta^*(r))^{-1}$. Thus two following products coincide by definition of A, B:

$$(\alpha, g)(\beta, h) = (\alpha \cdot g_*(\beta), \beta^*(g) \cdot h) = (\alpha A, 1),$$
$$(\gamma, r)(\delta, p) = (\gamma \cdot r_*(\delta), \delta^*(r) \cdot p) = (\gamma B, 1).$$

Putting $G = \{1\}$ we get that the condition (2) is necessary. □

Proof of Lemma 2.12. By Lemma A.1 there exist $r_1, r_2 \in \mathcal{C}G$ so that $t_1 g r_1 = r_2 = t_2 r_1$. By the definition of $\mathcal{C}G$ decompose $r_1 = T_1 h_1, r_2 = T_2 h_2$ with $T_i \in \mathsf{Arr}(\mathcal{C}), h_i \in G_*$, thus

$$t_1 g T_1 h_1 = T_2 h_2 = t_2 T_1 h_1.$$

It follows that $T_1^*(g) = 1$, by the injectivity we get that $g = 1$, $h_1 = h_2$. We deduce from above the equality $t_1 T_1 h_1 = t_2 T_1 h_1$. Canceling from right we get that $t_1 = t_2$. □

Proof of Lemma 3.4. Pullback of left (right) cancellable categories is left (right) cancellable. Thus, we need to prove that $\mathcal{E} \times_{\mathcal{B}} \mathcal{C}$ has common right multiples. Suppose that morphisms $f = (e_1, c_1)$ and $q = (e_2, c_2)$ have the same domain. Then by axiom O2, there exist morphisms r, s such that $c_1 r = c_2 s$ in $\mathcal{C}$. Let P_o be the bijection induced by p on the sets of objects of dom $e_1/\mathcal{E} \to$ dom $c_1/\mathcal{B}$. Put $\varphi := c_1 r$ to be the composition in $\mathcal{C}$.

$$\begin{tikzcd}
 & * \arrow[rr, dashed, "r_2"] \arrow[dd, dotted, mapsto] & & * \arrow[dd, dotted, mapsto] \\
* \arrow[ur, "e_1"] \arrow[rr, "e_2"] \arrow[dd, dotted, mapsto] & & * \arrow[ur, dashed, "s_2"'] \arrow[dd, dotted, mapsto] & \\
 & * \arrow[rr, dashed, "\exists\, r"] & & * \\
* \arrow[ur, "c_1"] \arrow[rr, "c_2"'] & & * \arrow[ur, dashed, "\exists\, s"'] &
\end{tikzcd}$$

Set $r_2 := e_1^{-1} P_o^{-1}(\varphi)$ and $s_2 := e_2^{-1} P_o^{-1}(\varphi)$. Then (r_2, r) is a correctly defined morphism in $\mathcal{E} \times_{\mathcal{B}} \mathcal{C}$ since $p(r_2) = p(e_1^{-1}) \cdot \varphi = c_1^{-1} c_1 r = r$ belongs to $\mathcal{C}$. By the same reasoning (s_2, s) is a correctly defined morphism in $\mathcal{E} \times_{\mathcal{B}} \mathcal{C}$. By definition $e_1 r_2 = e_2 s_2$, thus the morphisms (r_2, r) and (s_2, s) provide common right multiples. □

Proof of Lemma 3.5. Let us denote $\mathcal{Q} := \mathcal{C} \times_{\mathsf{Loc}\ \mathcal{C}} E$. By Lemma 3.4 the category $\mathcal{Q}$ is an Ore category, thus $\mathsf{Loc}\ \mathcal{Q}$ is its groupoid of fractions.

First, we prove **essential surjectivity** of the functor $\mathsf{Loc}\ \mathcal{Q} \to \mathcal{E}$. Note that the set of objects of $\mathcal{Q}$ is the same as the set of objects of $\mathsf{Loc}\ \mathcal{Q}$. Take an object $e \in \mathcal{E}$. Then $(e, p(e)) \in \mathcal{Q}$ is the preimage of e.

Next, we prove that the functor $\mathsf{Loc}\ \mathcal{Q} \to \mathcal{E}$ **is full**. For this we prove that any morphism in $\mathcal{E}$ can be represented as a fraction of two morphisms in $\mathcal{Q}$. Take a morphism $f\colon a \to b \in \mathcal{E}$. Since $\mathsf{Loc}\ \mathcal{C}$ is the groupoid of fractions of $\mathcal{C}$, $p(f)$ can be factored as $\varphi_1\varphi_2^{-1}$ for some $\varphi_1, \varphi_2 \in \mathcal{C}$.

$$\begin{tikzcd}
p(a) \arrow[rr, "p(f)"] \arrow[dr, "\varphi_1"'] & & p(b) \arrow[dl, "\varphi_2"] \\
& \bullet &
\end{tikzcd}$$

Let P_o be the bijection induced by p on the sets of objects of $a/\mathcal{E} \to p(a)/\mathsf{Loc}\ \mathcal{C}$.

Define $f_1 = P_o^{-1}(\varphi_1), f_2 := f^{-1}f_1$. Then $(f_1, p(f_1))$ and $(f_2, p(f_2))$ are correctly defined morphisms in $\mathcal{Q}$ and $f_1 f_2^{-1} = f$. Thus the functor is full.

Lastly, we need to prove that the functor $\mathsf{Loc}\ \mathcal{Q} \to \mathcal{E}$ **is faithful**. Since we work with groupoids we need to check that if image of some f is id_e then f itself is id_e for some e. Take $f = (g_\mathcal{C}, g_\mathcal{E})(h_\mathcal{C}, h_\mathcal{E})^{-1}$ in $\mathsf{Loc}\ \mathcal{Q}$ such that its image in $\mathcal{E}$ is id. It is easy to see that the image is $g_\mathcal{E} h_\mathcal{E}^{-1}$. We need to prove that it implies that $g_\mathcal{C} = h_\mathcal{C}$. We use Lemma A.1. Since $g_\mathcal{C} h_\mathcal{C}^{-1} = p(g_\mathcal{E} h_\mathcal{E}^{-1}) = \mathsf{id}$ we have a commutative diagram in $\mathcal{C}$ of the form

$$\begin{tikzcd}
& \bullet \arrow[d, "r_1"] & \\
\bullet \arrow[ur, "g_\mathcal{C}"] \arrow[dr, "\mathsf{id}"'] & \bullet & \bullet \arrow[ul, "h_\mathcal{C}"'] \arrow[dl, "\mathsf{id}"] \\
& \bullet \arrow[u, "r_2"'] &
\end{tikzcd}$$

That is, $g_\mathcal{C} r_1 = r_2$ and $h_\mathcal{C} r_1 = r_2$ which imply by cancellability that $g_\mathcal{C} = h_\mathcal{C}$

□

Proof of Lemma 3.6. First, note that the category $\mathcal{Q}$ can also be presented as the pullback $\mathcal{C} \to \mathsf{Loc}\ (\mathcal{C}\ G) \leftarrow \mathcal{E}$. This follows from a well-known categorical result (also known as the pasting lemma), that in a commutative diagram as below, where the right square is a pullback, the left square is a pullback if and only if the outer rectangle is a pullback.

$$\begin{tikzcd}
\bullet \arrow[r] \arrow[d] & \bullet \arrow[r] \arrow[d] \arrow[dr, phantom, very near start, "\lrcorner"] & \bullet \arrow[d] \\
\bullet \arrow[r] & \bullet \arrow[r] & \bullet
\end{tikzcd}$$

Let us denote by $p\colon \mathcal{E} \to \mathsf{Loc}\ \ \mathcal{C}G$ the covering map on the right. If $(f_\mathcal{C}, f_\mathcal{E})$ is an isomorphism with inverse $(g_\mathcal{C}, g_\mathcal{E})$ then $f = g = \mathsf{id}_e$ since $\mathcal{C}$ is gaunt. Then, $f_\mathcal{E}$ is also an id morphism by the covering properties. Thus, $\mathcal{Q}$ is gaunt.

Now we prove that each morphism in X factors uniquely as a morphism in $\mathcal{Q}$ followed by an isomorphism. Take $(f_{\mathcal{C}G}, f_\mathcal{E})$ a morphim in X. To prove the existence of such decomposition, represent $f_{\mathcal{C}G} = f_\mathcal{C} f_G$, consider P_o to be the bijection induced by p on the sets of objects of dom $f_\mathcal{E}/\mathcal{E} \to$ dom $f_\mathcal{C}/\mathsf{Loc}\ \mathcal{C}$ and take $f_\mathcal{E}^1 = P_o^{-1}(f_\mathcal{C})$ and $f_\mathcal{E}^2 = (f_\mathcal{E}^1)^{-1} f_\mathcal{E}$. Then obviously $(f_\mathcal{C}, f_\mathcal{E}^1)(f_G, f_\mathcal{E}^2)$ is the desired decomposition. To prove the uniqueness suppose that $(f_1, e_1)(f_2, e_2) = (g_1, t_1)(g_2, t_2)$ for some $(f_1, e_1), (g_1, t_1) \in \mathcal{Q}$ and $(f_2, e_2), (g_2, t_2)$ isomorphisms. Then by uniqueness of decomposition in $\mathcal{C}G$ we have $f_1 = g_1, f_2 = g_2$. By covering properties, there is a unique morphism $\psi \in \mathcal{E}$ with dom $\psi =$ dom e_1 such that $p(\psi) = f_1$. Thus, $e_1 = t_1 = \psi$, thus the decomposition is unique.

□

Lemma A.2. *Let $\mathcal{C}$ be a category with wide subcategory $\mathcal{W} \subseteq \mathcal{C}$ such that each morphism in $\mathcal{C}$ can be uniquely decomposed as a morphism in $\mathcal{W}$ followed by an isomorphism. Then, $\mathcal{C}$ is equivalent to $\mathcal{W}G$ for some crossed $\mathcal{W}$ group G.*

Proof. We prove that the skeleton of $\mathcal{C}$ is a category of a crossed $\mathcal{W}$ group. It is trivial that $\mathcal{W}$ is gaunt. Take a set of representatives $\{g_i\}_{i\in\mathcal{I}}$ for each isomorphism class. Denote as $[g_i]$ the set of objects of $\mathcal{C}$ that are isomorphic to g_i. For each $h \in [g_i]$ choose an isomorphism $\alpha_i^h : h \to g_i$. This defines a functor $\mathcal{C} \to \mathcal{B}$ to the full subcategory spanned by g_i's.

□

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